Open problems:
Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Completeness of the Lie algebra.** *Research.* Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. $\mathcal{S}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^m)$ **iff** $n = m$**.** Stated in Example 3 and argued only by "no linear transformation can change the number of defining observables", but the number of defining observables is not an invariant. With the two notions of isomorphism in the formalization, the claim holds for Lie isomorphisms, by the dimension $2n+1$ of the Lie algebra of X^i, P_i, I, and fails for unitary equivalences, by the Hermite expansion. Neither argument is formalized; the first needs the linear independence of X^i, P_i, I on $\mathcal{S}(\mathbb{R}^n)$. (Brief, Example 3 and Remark; Log, "Differences", third- and second-to-last items.)

3. **Lie algebras determine physical systems.** *Research.* "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

4. **Ensembles on a quantum vector space.** *Research.* QSTO:V uses density operators ρ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over V is defined. In particular, whether the ensembles induce on the pure states the quotient topology of V, under which the pure states are second countable exactly when V is, is open. (Brief, Sec. 3, QSTO:V; Sec. 5.)

5. **Field theories.** *Research.* "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

6. **Which observables generate unitary groups.** *Research.* Since symmetric and self-adjoint coincide on V, "not all self-adjoint operators generate a unitary transformation", and "to understand which ones the integrability of the system must be checked". A characterization,

analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

From the formalization

None. Definition 1 and Examples 3 (in n dimensions) and 4 are checked in full.